

The binary fraction of LMC RR Lyrae from 34 years of pulsation timing

Project `rrl_lmc_binary` — working note

7 October 2026 (pipeline v4; code and data products:
https://github.com/vasilybelokurov/rrl_lmc_binary)

Abstract

A companion makes an RR Lyrae star move towards and away from us, so its pulsations arrive alternately early and late by the light-travel time across the orbit: $\sim 5\text{--}30$ min for companions of $0.4\text{--}1.5 M_{\odot}$ on $3\text{--}27$ -yr orbits. We measure the pulsation arrival time once per observing season for 17,490 fundamental-mode RR Lyrae (RRab) in the LMC, using MACHO (1992–1999), OGLE-II/III (1997–2009) and OGLE-IV photometry extended to 2026 by the OGLE team. For the 6,612 stars with MACHO data the record spans 34 years. Most stars show slow timing variations, but these are explained by the stars’ own random period wander: simulations with realistic intrinsic timing noise reproduce the distribution of the orbit-search statistic and predict 112 orbit-like candidates where 93 are observed. There is therefore no detectable binary population. Injecting Keplerian orbits into the real stars’ timing data shows that a $0.4\text{--}1.5 M_{\odot}$ companion on a $3\text{--}27$ -yr orbit is flagged with probability 32%. Counting *every* candidate as a binary, the binary fraction for such companions is $f < 5.3\%$ (95%), and below 4.8% for $8\text{--}27$ -yr orbits. One candidate, OGLE-LMC-RRLYR-13854, passed a blind test: its orbit, fitted to data up to 2016 and frozen before the new data arrived, correctly predicted the following decade. Timing alone cannot prove binarity, because some RR Lyrae show strictly periodic intrinsic timing modulation; radial velocities are needed to confirm the best candidates.

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1 Summary

- **Result.** Fewer than 5.3% of LMC RRab have a $0.4\text{--}1.5 M_{\odot}$ companion on a 1,000–10,000-d orbit (95% confidence; [section 7](#), [table 1](#)). For $0.15\text{--}0.4 M_{\odot}$ companions on 3,000–10,000-d orbits: $f < 9.6\%$. Lower-mass or shorter-period companions are not constrained.
- **Why the fraction cannot be higher.** A binary in this range would be flagged by our search with probability 32%, measured by adding orbits to the real stars’ own timing data. If more than 5.3% of the 6,612 stars were such binaries, more than 110 of them would be flagged; we flag only 93 stars in total, even counting all of them as binaries.
- **Most RR Lyrae show no sign of a binary.** The orbit-like signals in the data are what the stars’ intrinsic timing noise produces: the number of candidates (93) is consistent with, and slightly below, the number expected from noise alone (112; [section 5](#)). The data are therefore consistent with a binary fraction of zero; they give no lower limit.
- **Candidates.** A short list of 12 stars ([section 8](#)) passes physical, amplitude-stability and prediction checks; 13854 and 15158 passed a blind prediction test. Their predicted radial-velocity semi-amplitudes are $3\text{--}12 \text{ km s}^{-1}$.

2 Data

- **OGLE.** *I*-band light curves of the OGLE Collection of Variable Stars: OGLE-III [\[8\]](#) including OGLE-II epochs from 1997, and OGLE-IV [\[9\]](#). The OGLE team (I. Soszyński, priv. comm., 6 Oct 2026) provided OGLE-IV photometry for 2010–2026 for 38,476 LMC RR Lyrae, including the 2022–2024 high-cadence campaign on the central fields [\[6\]](#). The central fields (LMC502, 503, 509, 510, 511, 516) were re-reduced after 2016 (per-epoch differences of a few mmag against errors of ~ 40 mmag), and the raw extraction contains rare outliers ($\sim 0.05\%$ of epochs beyond 5σ); both are handled by per-segment zero points, per-season offsets and clipping.
- **MACHO.** *B* and *R* light curves (1992–1999) from the MACHO archive, for 6,612 of the stars.
- **Sample.** 17,490 RRab with OGLE-III and OGLE-IV data. The binary-fraction measurement uses the 6,612 stars with MACHO data (34-yr baseline, $\sim 30\text{--}40$ seasons); they lie in the central LMC fields covered by MACHO.

3 Measurement: the arrival time of the pulsation

For each star and band we fit one Fourier template (8 harmonics) to all epochs, with a separate zero point per survey segment, and measure in each observing season j a time shift τ_j of the template (the delay), together with a mean-magnitude offset and an amplitude scale α_j :

$$m(t) = z_s + \Delta m_j + \alpha_j T\left(\frac{t - \tau_j - T_0}{P}\right). \quad (1)$$

The delay error is the Fisher error scaled by $\sqrt{\chi^2_{\nu}}$; seasons are clipped for outliers and dropped when the delay depends on the clipping or the χ^2 surface has two comparable minima (1.4% of seasons). The sequence τ_j is the $O-C$ diagram (figure 1). A slow, constant change of the pulsation period gives a parabola; it is fitted and removed in every model. Shared year-to-year timing systematics (up to ~ 130 s in MACHO, $\lesssim 70$ s in OGLE) are estimated from the whole sample and subtracted. The MACHO–OGLE timing offset per band follows a calibrated period relation (scatter ~ 300 s). The typical season delay error is 220 s (100–500 s).

Signal of a companion. The light-travel delay of an RR Lyrae of mass M_1 with a companion M_2 on an orbit of period P_{orb} has semi-amplitude

$$A = \frac{a_1 \sin i}{c} = \frac{\sin i}{c} \frac{M_2}{M_1 + M_2} \left[\frac{G(M_1 + M_2)P_{\text{orb}}^2}{4\pi^2} \right]^{1/3}, \quad (2)$$

e.g. $A = 505$ s for $M_1 = 0.65 M_{\odot}$, $M_2 = 0.6 M_{\odot}$, $P_{\text{orb}} = 1000$ d and 1050 s at 3000 d (edge-on).

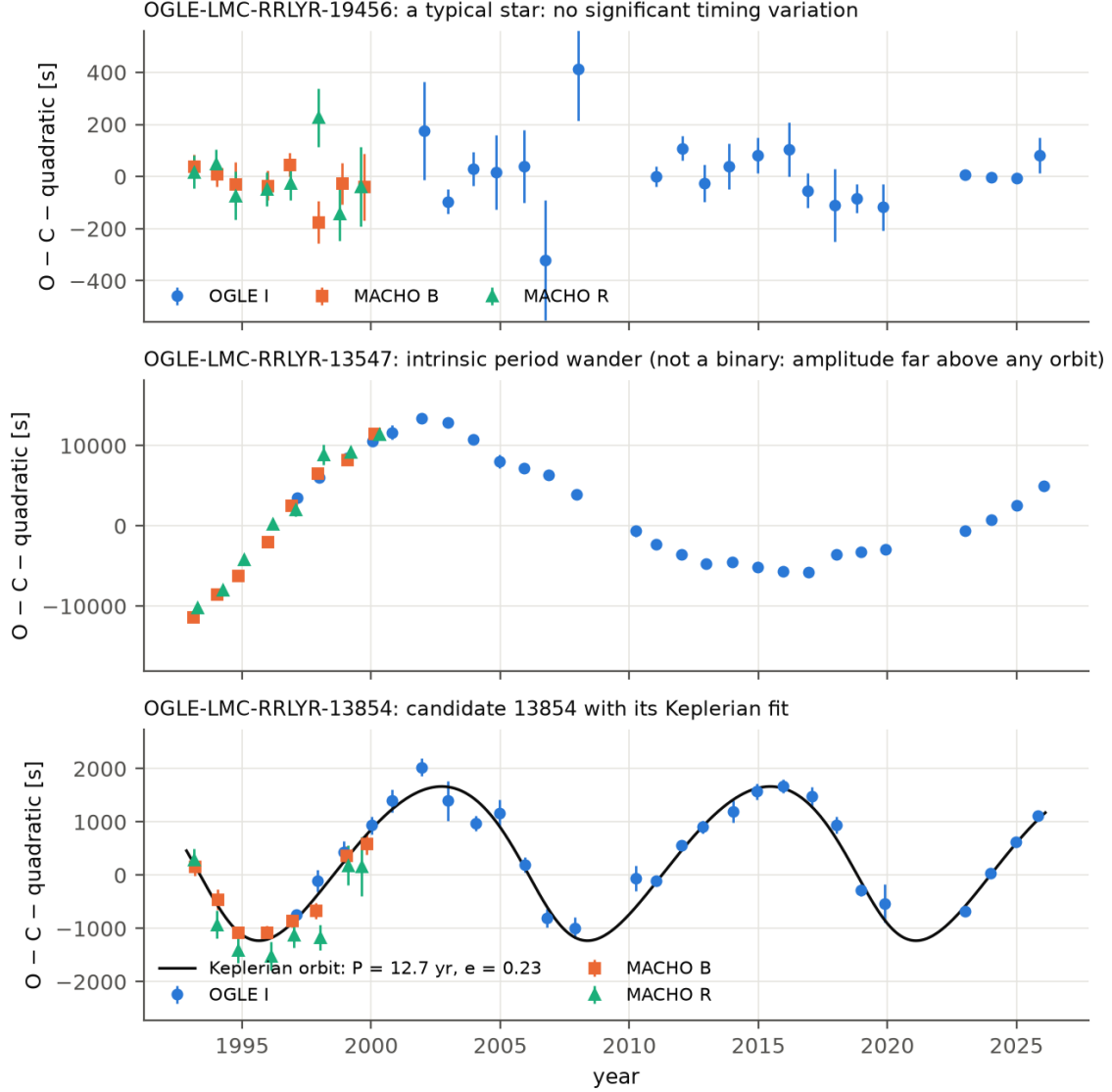


Figure 1: Season delays ($O - C$, a fitted parabola removed) for three stars. Top: a typical star; the delays are consistent with the errors. Middle: a star with large intrinsic period wander (12,000 s, far above any orbit). Bottom: the candidate 13854 with its Keplerian fit ($P = 12.7 \text{ yr}$, $e = 0.23$); MACHO *B*, *R* and OGLE follow the same curve over 2.6 cycles.

4 The search

For each star we fit, by maximum likelihood, the parabola plus band offsets (null model) and the same plus a sinusoid of period P_{orb} on a grid from 400 d to twice the baseline (orbit model), with a white jitter term in both. The statistic is $D = 2 \max_{P_{\text{orb}}} [\ln L_{\text{orbit}} - \ln L_{\text{null}}]$. A star is a candidate if it passes five cuts: (1) $D > 40$; (2) a steady pulsation amplitude, $\chi^2_\nu(\alpha_j) < 2$ (an orbit does not change the light curve; Blazhko-type modulation does); (3) orbit amplitude larger than 3 season errors; (4) at least 1.5 cycles in the baseline; (5) amplitude below that of an edge-on $2 M_\odot$ companion at the fitted period (larger timing variations cannot be orbits). 93 of the 6,612 stars pass.

5 Most RR Lyrae show no sign of a binary

Intrinsic timing noise. RR Lyrae change their pulsation period irregularly. Fitting the delays of each star with a smooth random-wander model gives its intrinsic noise; the noise is common to all bands (correlation of B and R residuals 0.58), so it is a property of the star, not of the photometry. We simulate light curves of real stars on their real epochs and with their errors, with timing noise drawn from the real stars' noise fits (candidates excluded), and pass them through the identical pipeline.

The statistic D . With measurement noise only, no simulated star reaches $D = 40$ (figure 2). With realistic intrinsic timing noise the simulated distribution follows the real one closely over three decades; injected orbits move stars to large D , but the real stars do not show such an excess.

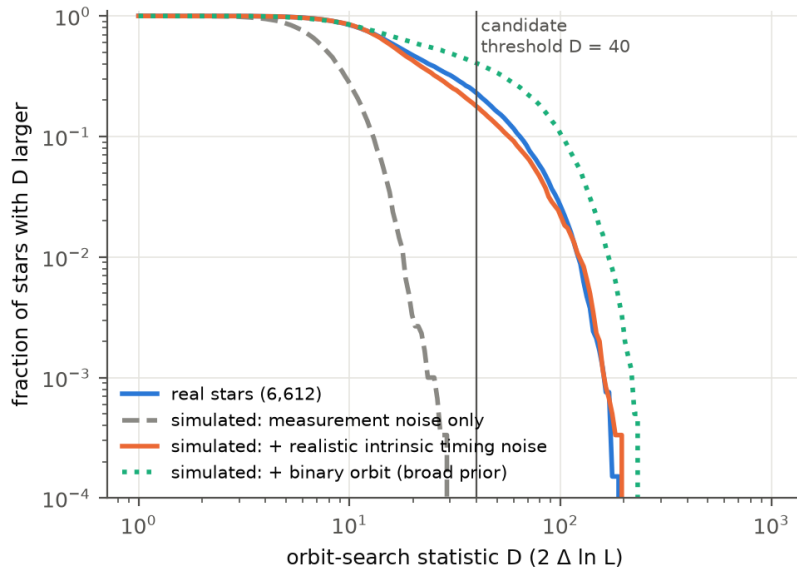


Figure 2: Fraction of stars with an orbit-search statistic above D : real stars (blue) and simulated stars of the same sample with measurement noise only (grey), with realistic intrinsic timing noise (orange) and with injected orbits (green; broad prior: $P = 300\text{--}10^4$ d, $M_2 = 0.05\text{--}1.5 M_\odot$, random inclination).

Candidates vs noise. At each cut the number of real stars is close to the number expected from intrinsic noise alone (figure 3). At $D > 40$ the real stars exceed the noise model by 30% (1,519 vs 1,172), an imperfection of the noise model at moderate D , but after the cuts that select orbit-like

signals the real count is *below* the expectation: 93 observed vs 112 expected. There is no excess of orbit-like signals.

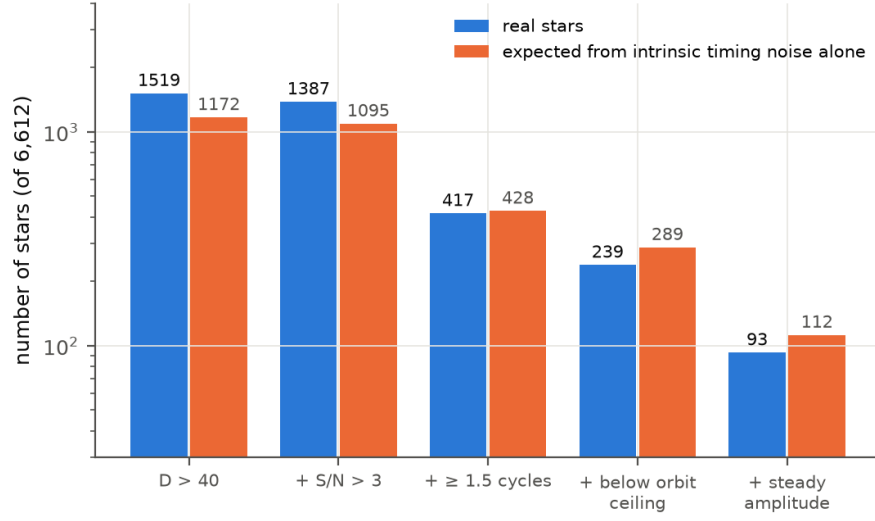


Figure 3: Number of stars passing the successive cuts: real (blue) and expected from intrinsic timing noise alone (orange; the simulated noise carries no amplitude variability, so the last step uses the real amplitude-stability pass rate of stars passing the other cuts).

Amplitudes. The best-fit sinusoids of most stars are small (figure 4): typically 100–300 s at periods of 1–3 yr, set by the noise. The candidates lie at 300–3000 s and 5–20 yr, where the intrinsic wander is largest.

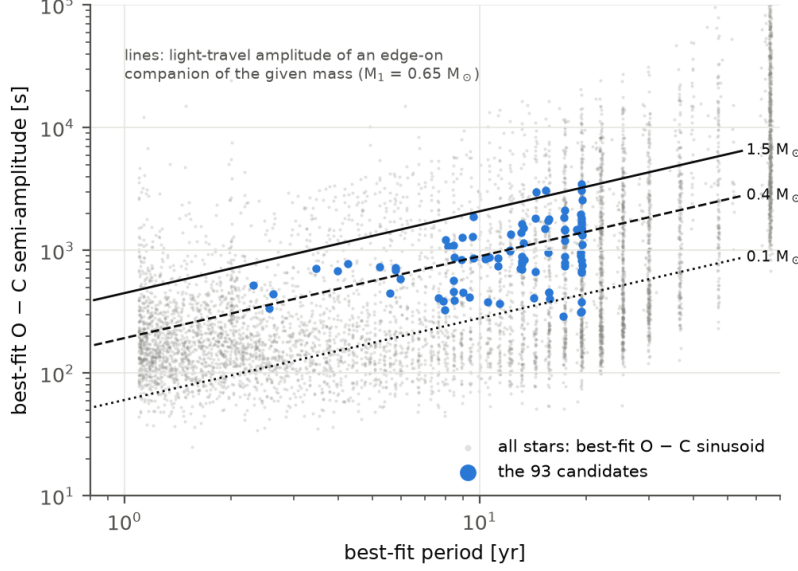


Figure 4: Best-fit $O-C$ semi-amplitude and period of every star (grey) and of the 93 candidates (blue), with the light-travel amplitude of an edge-on companion of 0.1, 0.4 and $1.5 M_{\odot}$. The vertical stripes at long periods are the period grid; values above the $1.5 M_{\odot}$ line and beyond ~ 20 yr are intrinsic wander or trends.

6 Detection efficiency

The probability that a binary is flagged is measured by injection into the *real* stars: for each of the 6,612 stars, two Keplerian orbits are drawn (P_{orb} log-uniform 300– 10^4 d, M_2 log-uniform 0.05 – $1.5 M_{\odot}$, $M_1 = 0.65 M_{\odot}$, isotropic inclination, half circular and half with e uniform in 0 – 0.7) and their light-travel delay is added to the star’s measured season delays in all bands; the search and the cuts are then rerun unchanged. The injected binaries therefore carry each star’s real intrinsic timing noise, real amplitude behaviour (so Blazhko-type stars are rejected exactly as in the data), cadence and errors. The orbit is added at the season epochs, neglecting the smearing of the delay within a season (amplitude loss $\lesssim 10\%$ at 1000 d, $\lesssim 1\%$ at 3000 d); injecting the same orbits at the season level into simulated stars without timing noise gives an efficiency 0.586 vs 0.568 for the light-curve-level injection (0.4 – $1.5 M_{\odot}$, 1,000–10,000 d) and 0.131 vs 0.101 at 300–1,000 d. The real-star efficiencies are therefore multiplied by this ratio in each bin (capped at 1; conservative).

Figure 5 shows the result. For 0.4 – $1.5 M_{\odot}$ companions on 1,000–10,000-d orbits the efficiency is 32% (32.6% before the smearing correction; the earlier estimate from simulated stars without intrinsic timing noise, multiplied by the fraction of real stars with a steady amplitude, was 34.6%, close to the direct measurement); it falls to 7% at 300–1,000 d (small amplitudes, season smearing) and is 17% for 0.15 – $0.4 M_{\odot}$ at 3,000–10,000 d. Binaries are missed because their orbit is seen nearly face-on, because the host star varies in amplitude (Blazhko type; the amplitude cut rejects it), or because the host’s own timing noise distorts the signal.

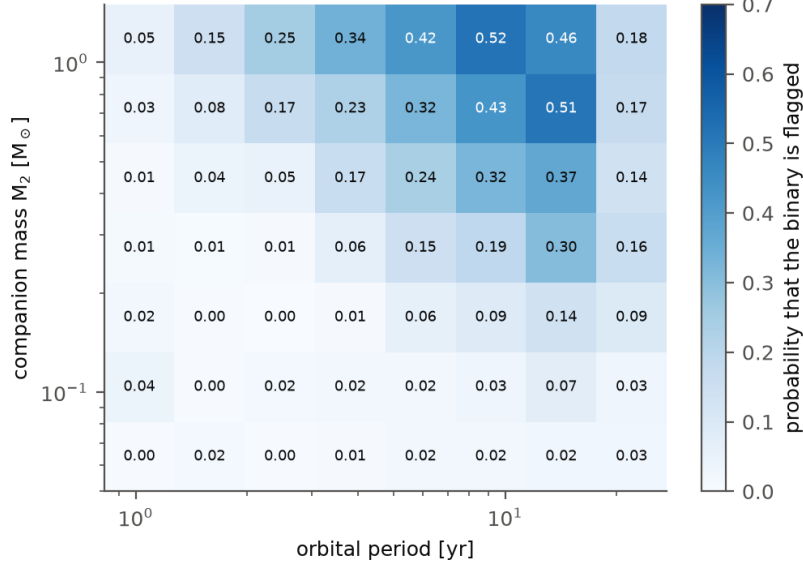


Figure 5: Probability that a binary is flagged as a candidate, from orbits injected into the real stars’ timing data, in bins of orbital period and companion mass (averaged over inclination and eccentricity; before the season-smearing correction, which lowers the values by 30% below 1000 d and $\leq 6\%$ above). The values of 0.01–0.04 for low-mass companions are mostly stars that are flagged because of their own timing noise.

7 Upper limit on the binary fraction

Argument. Let a fraction f of the $N = 6,612$ stars have a companion from the injected population of a given bin. The expected number of such binaries that are flagged is $fN\varepsilon$. We flag $k = 93$ stars in total. Counting *every* one of them as a binary (no subtraction of noise candidates) we require $fN\varepsilon \leq k_{95}$, where $k_{95} = 110$ is the 95% upper limit on a Poisson mean given k observed, so

$$f < \frac{k_{95}}{N\varepsilon}. \quad (3)$$

Binaries that are not flagged are allowed by construction: they are accounted for by ε . The argument does not depend on the model of the intrinsic noise; it only needs the efficiency, which is measured on the real stars.

Result. Table 1 and figure 6. If the 112 candidates expected from noise are subtracted, the limit becomes much tighter (0.6%; 2.6% if the noise expectation is halved), but this depends on the noise model and we do not adopt it.

Assumptions. (i) The limit refers to the injected population within each bin (log-uniform period and mass, isotropic orientation, $e < 0.7$); a population concentrated where the efficiency is lower gives a weaker limit. (ii) Binarity is assumed independent of the host’s amplitude behaviour and intrinsic timing noise (the injections place orbits into randomly chosen real stars). (iii) The sample is the MACHO-covered central LMC. (iv) Real stars could already contain undetected binaries; adding an orbit to such a star can only increase its detectability, a negligible effect at $f \lesssim 5\%$.

Table 1: 95% upper limits on the binary fraction of LMC RRAb (6,612 stars with MACHO+OGLE data, 1992–2026). ε : probability that a binary of the bin is flagged (from injections into the real stars; 68% range from the number of injections). $f_{\max}$: counting all candidates as binaries. $f_{\max}^{\text{bkg}}$: noise candidates subtracted (model dependent).

$M_2 [M_\odot]$	$P_{\text{orb}} [\text{d}]$	injections	ε (68%)	$f_{\max}$	$f_{\max}^{\text{bkg}}$
0.4–1.5	300–1000	1766	0.072 (0.067–0.078)	23.1%	2.6%
0.4–1.5	1000–3000	1563	0.280 (0.269–0.292)	6.0%	0.7%
0.4–1.5	3000–10000	1807	0.347 (0.335–0.358)	4.8%	0.5%
0.4–1.5	1000–10000	3370	0.316 (0.308–0.324)	5.3%	0.6%
0.15–0.4	300–1000	1311	0.003 (0.002–0.004)	–	–
0.15–0.4	1000–3000	1177	0.090 (0.082–0.099)	18.6%	2.1%
0.15–0.4	3000–10000	1373	0.175 (0.165–0.185)	9.6%	1.1%
0.15–0.4	1000–10000	2550	0.140 (0.134–0.148)	11.9%	1.3%
0.05–0.15	300–1000	1451	0.013 (0.010–0.017)	–	–
0.05–0.15	1000–3000	1337	0.020 (0.016–0.025)	82.8%	9.4%
0.05–0.15	3000–10000	1439	0.042 (0.036–0.048)	40.1%	4.5%
0.05–0.15	1000–10000	2776	0.031 (0.028–0.035)	53.3%	6.0%

“–”: no useful limit ($f_{\max} > 100\%$).

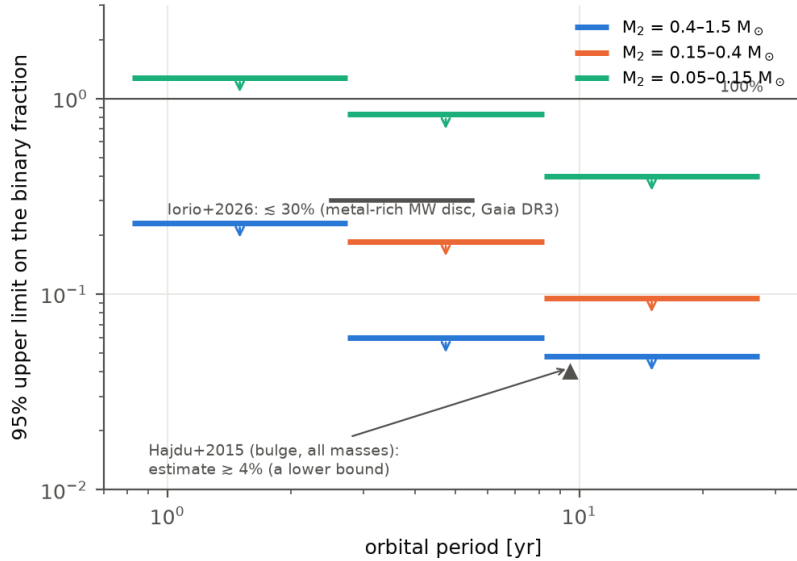


Figure 6: 95% upper limit on the binary fraction vs orbital period for three companion-mass ranges. Literature values are shown for orientation; they refer to different methods and populations (section 10).

8 Individual candidates and the blind test

On 1 October 2026, before the OGLE data after 2016 were available, we froze orbit predictions for 2016–2026 for 28 candidates from the data up to 2016, with the predictive uncertainty from the stars’ red-noise model. The new data test them (figure 7): 2 predictions were confirmed (13854, 15158), 20 failed and 5 were inconclusive (one star had no new data). A failed prediction shows that the frozen best-fit orbit was wrong, not that the star is single; a passed one is evidence for a coherent, orbit-like signal under the adopted noise model.

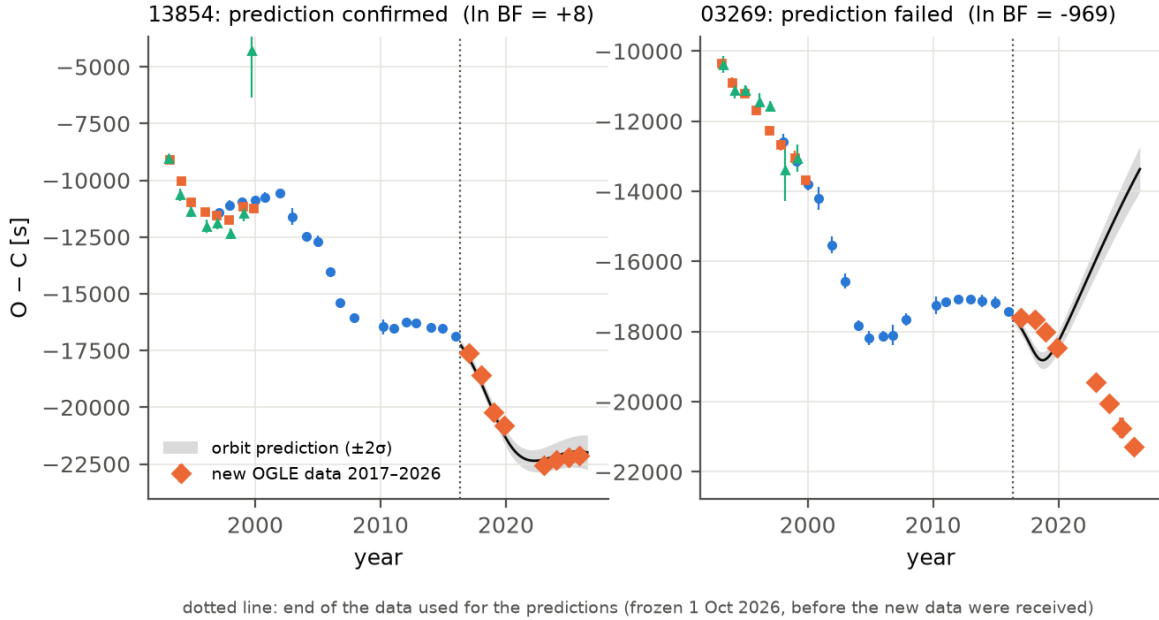


Figure 7: Blind test. Points to the dotted line: data used for the prediction (MACHO B , R ; OGLE I). Grey band: frozen orbit prediction ($\pm 2\sigma$). Orange: OGLE seasons 2017–2026. Left: 13854, confirmed. Right: 03269, failed.

A follow-up short list (12 stars) requires: a physically possible orbit (amplitude below an edge-on $2 M_{\odot}$ companion), at least two orbital cycles in 34 years, a steady pulsation amplitude, and a successful prediction of the 2017–2026 data (blind for 13854 and 15158; not blind for stars first selected with the new data). Table 2 lists them.

Table 2: Follow-up short list. P , e : Keplerian fit to 1992–2026. $M_{2,\min}$: minimum companion mass ($M_1 = 0.65 M_\odot$, edge-on). K_1 : predicted radial-velocity semi-amplitude of the RR Lyrae.

OGLE-LMC-RRLYR-	P [yr]	e	$M_{2,\min} [M_\odot]$	K_1 [km s $^{-1}$]	prediction	eccentric	data
15158	8.5	0.57	0.76	10.7	blind	yes	MACHO+OGLE
13854	12.7	0.23	0.64	7.1	blind	no	MACHO+OGLE
13083	7.9	0.46	0.45	7.1	split	yes	OGLE
00604	6.9	0.16	0.53	7.5	split	no	OGLE
08275	7.8	0.28	0.16	2.9	split	no	MACHO+OGLE
00723	5.4	0.26	0.87	11.5	split	no	OGLE
11047	12.8	0.34	0.42	5.4	split	yes	MACHO+OGLE
04633	13.5	0.69	0.64	9.3	split	yes	MACHO+OGLE
03700	5.8	0.22	0.51	7.9	split	no	MACHO+OGLE
08101	13.5	0.52	0.88	9.6	split	yes	MACHO+OGLE
18547	12.8	0.51	0.26	4.2	split	no	MACHO+OGLE
16301	13.6	0.75	0.44	7.7	split	no	MACHO+OGLE

9 What timing can and cannot establish

- A circular orbit produces a sinusoidal delay. A strictly periodic intrinsic phase modulation produces the same signal. In simulations on the real cadences, circular orbits cannot be separated from such modulation (power $\leq 13\%$ at 1% false alarm); strongly eccentric orbits ($e = 0.5$, amplitude 8 season errors) can (93–97%), because the Keplerian waveform is distinctive.
- Such modulation exists: about 2.4% of RR Lyrae show a periodic timing component beyond random wander, and some of these are strictly coherent with amplitudes too large for any orbit. About 90% of the coherent intrinsic modulators also vary in pulsation amplitude, which an orbit does not do; this is why the amplitude-stability cut is central.
- Consequently individual candidates remain candidates until radial velocities (semi-amplitudes 3–12 km s $^{-1}$ over years) or Gaia epoch astrometry/photometry confirm them. The *upper limit* is not affected: it counts all candidates as binaries.

10 Comparison with the literature

- Hajdu et al. [2] found 20 probable light-travel-time binaries among 1,952 OGLE-III bulge RRab ($\sim 1\%$), doubled this for Blazhko stars and again for missed orbits, and concluded that $\gtrsim 4\%$ of RR Lyrae are in binaries. This is a lower estimate that assumes the candidates are real and covers all companion masses. Their raw candidate rate is close to ours ($93/6,612 = 1.4\%$); our simulations show that such a rate is produced by intrinsic timing noise, so the candidates alone do not establish a lower bound.
- Hajdu et al. [3] presented 87 bulge candidates with $P_{\text{orb}} > 1000$ d and companion masses clustering near 0.6, 0.2 and $0.067 M_\odot$; none has been confirmed or refuted since [7].
- Kervella et al. [5] found proper-motion anomalies (Hipparcos vs Gaia DR2) in 13 of 198 nearby RR Lyrae, plus 61 candidates, and concluded that at least 7% are binaries. That method is sensitive to different orbits and companion masses and probes the solar neighbourhood.

- Iorio et al. [4] found no genuine RR Lyrae in the Gaia DR3 astrometric binary catalogues; for metal-rich disc RR Lyrae with predicted orbits of $\sim 900\text{--}2000$ d, their fiducial models give an upper limit of $\approx 30\%$. The LMC sample is mostly metal-poor, so our limit does not test that binary-evolution channel [1].

11 Reproducibility

Scripts (repository): `level1_real.py`, `level2.py` (timing and search; run with `-ogle4 extended`); `level1_sims.py` (light-curve simulations); `inject_real.py` (efficiency); `upper_limit_v4.py` (limits, `results/partB_v4/upper_limits.csv`); `shortlist_v4.py`; `test_frozen_predictions.py` (blind test; frozen inputs under git tag `predictions-2026-10-01`); `plot_binary_fraction.py` (figures). Tests: `pytest`. Details and all intermediate numbers: `JOURNAL.md`.

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